Subwavelength Lateral Heterogeneity: Diffraction Processing and Lateral Tuning Theory

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1 Processing Workflow: Isolating and Analyzing Scattering

To analyze the weak diffracted field generated by subwavelength lateral variations, the dominant coherent reflection must be systematically removed.

1. **Background Removal / Coherent Subtraction:** Obtain the isolated scattered field $u_{scat}(x,t)$ by subtracting the mean coherent field $u_{avg}(x,t)$ from the full data $u_{full}(x,t)$.
   - *Method A (Synthetic Reference):* Subtract a forward-modeled homogeneous equivalent fracture.
   - *Method B (Data-Driven):* Apply a robust lateral low-pass filter (or spatial moving average) to estimate $u_{avg}(x,t)$, then compute $u_{scat}(x,t) = u_{full}(x,t) - u_{avg}(x,t)$. (We often use Truncated SVD for this).
2. **F-K Filtering:** Transform u_{scat} to the $f-k$ domain. Apply a surgical mute or high-pass filter to k_x to remove residual near-dc ($k_x \approx 0$) energy left over from imperfect background subtraction.
3. **Diffraction Migration:** Perform prestack depth migration exclusively on u_{scat}. Rather than continuous flat events, this will image point-like diffractors at the phase boundaries of the internal blocks.
4. **Multi-Band Spectral Analysis:** Perform the migration using both low-frequency and high-frequency bandpass filters. Low frequencies map the macro-structure; high frequencies extract the internal scattering variations.

2 Theoretical Framework: Fracture Grating & Bragg Scattering

We model the fracture as a thin layer with a mean background property and lateral perturbations. The internal structural variations act as a subwavelength spatial grating.

2.1 Model Assumptions

Let the horizontal spatial perturbation be periodic or quasi-periodic:

$$\delta m(x) = A\cos(K_0 x) \tag{1}$$

where $K_0 = 2\pi/d$ is the spatial wavenumber of the blocks, d is the block width, and A is the contrast amplitude.

2.2 Bragg Scattering & Band-Limited Observation

When an incident wave $U_{inc}(x,\omega)$ with wavenumber $k_{x,inc}$ interacts with this grating, the scattered field is generated via Bragg scattering:

$$k_{x,scat} = k_{x,inc} \pm K_0 \tag{2}$$

For the scattered energy to reach the surface receivers, it must propagate rather than decay exponentially. Therefore, the scattered wavenumber must lie within the propagating cone:

$$|k_{x,scat}| \leq \frac{\omega}{v} = k_{medium} \tag{3}$$

This imposes a fundamental bandpass filter on the observable internal spectrum. High spatial-frequency variations (very small d) generate large K_0. If $K_0 > k_{medium}$, the scattered waves are purely evanescent and undetectable in the far-field. The observed $f-k$ spectrum $U(k_x, \omega)$ is therefore the true lateral spectrum $S(k_x)$ truncated by the host medium's propagation capabilities:

$$|U(k_x, \omega)|^2 \approx |G(\omega, k_x)|^2 S(k_x) \tag{4}$$

where G accounts for acquisition geometry, wavelet spectrum, and wave propagation physics.

3 Proposed "Lateral Tuning Law"

Analogous to the vertical thin-bed tuning effect ($E \propto \sin(k_z \Delta z)$), we define a lateral tuning parameter:

$$\eta = \frac{d}{\lambda} \tag{5}$$

where λ is the dominant wavelength in the background medium.

The diffracted energy $E_{diff}(\omega)$ follows a lateral tuning relationship:

$$E_{diff}(\omega) \propto |\delta m(K_0)|^2 \cdot C(\eta) \tag{6}$$

where $C(\eta)$ is the diffraction coupling efficiency:

- $\eta \gg 1$ **(Geometric Regime):** Blocks are large enough to act as independent specular reflectors. Standard resolution applies.
- $\eta \approx 0.5 - 1.0$ **(Resonant Tuning Regime):** The spatial period perfectly matches the horizontal wavenumber limits of the propagating wave. This yields maximum constructive interference (Bragg resonance) back to the receivers. $C(\eta)$ approaches its peak.
- $\eta < 0.5$ **(Evanescent/Subwavelength Regime):** K_0 exceeds the radiation cone. $C(\eta)$ drops off exponentially as the scattering becomes primarily evanescent. The layer effectively homogenizes.

4 Practical Measurable Quantities

To invert for subwavelength structures, parameterize the data using the following metrics extracted from u_{scat}:

1. **F-K Ridge Peaks (K_{peak}):** Search the $f-k$ spectrum for maximum energy ridges. If the structure is periodic, the peak k_x directly estimates K_0, yielding $d = 2\pi/K_0$.
2. **Frequency-Dependent Amplitude Decay:** Calculate total diffracted energy integrating over k_x for various frequency bins.
3. **Migrated Diffraction Focusing Energy:** Compute the localized energy of the migrated image $I(x, z)$ at the known fracture depth to invert for the perturbation contrast A.